Portland State University

PH 322 Computational Physics
Spring 2026

Instructor Information

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Class meetings: Tu/Th 11am-12:40pm, FMH B155
Office hours: Wednesdays 2-4 pm in person (SRTC 470) or through Zoom https://pdx.zoom.us/j/6362558378
Contact: Please email me at cbuten@pdx.edu instead of Canvas Mail

Course Overview

The purpose of this course is to learn numerical methods in order to solve physics problems using computers. We will learn techniques to solve a number of problems that are frequently encountered in physics including integration, differentiation, Fourier series, ordinary and partial differential equations. The course does not assume previous computer programming experience. We will spend the first two weeks learning the basics of Python.

Learning Outcomes

- Write, test, and debug Python programs that implement commonly used numerical methods in physics.
- Apply numerical techniques including integration, differentiation, root-finding, differential equation solving, and Fourier analysis to investigate physical problems.
- Explain the mathematical foundations of numerical methods, including Taylor series, Fourier series, discretization, approximation, and truncation error.
- Evaluate the accuracy of computational results by analyzing numerical and approximation error.
- Visualize and interpret computational results using graphs and other visualizations.
- Apply Python and numerical techniques to a novel problem or research project and communicate results through written and visual reports.

Textbook: *Computational Physics* by Mark Newman. There is a companion website that contains Python code and data files for textbook examples and other information you might find useful. The text is available at the PSU Bookstore and through online retailers. Purchase the book as soon as possible. The author provides full PDFs of the first few chapters on his website. The second edition is recommended.

Python: We will use the Python programming language throughout the course. All homework code must be written in Python. Python is freely available to download and install on Windows, MacOS, and Linux operating systems. There are many different distributions of Python that come with different Python consoles, text editors and other bells and whistles. These will probably all work fine since they all contain the same core Python language, but some make your programming experience better. I personally use the Anaconda distribution and the

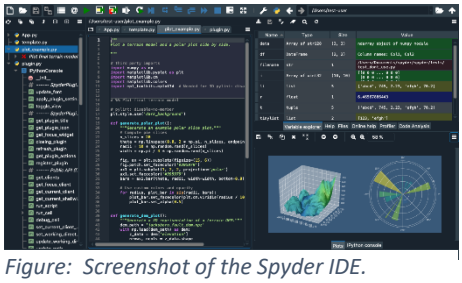

Figure: Screenshot of the Spyder IDE.

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Spyder integrated development environment (IDE), which I recommend. The Spyder IDE includes the IPython console, a text editor, a debugger and a variable explorer which is useful to debug your code. You can download Anaconda from https://www.anaconda.com/products/individual. I put additional information about installing Anaconda and Spyder on Canvas.

Google Colab**:** If you don't want to install your own distribution of Python, you can program in Python through the cloud using Google CoLab. CoLab is a web-based environment where you can write and run Python code in a notebook through your browser. A Colab notebook looks a lot like a Jupyter notebook: it has text cells and code cells mixed together. You can write explanations in one cell and run Python in the next. Conveniently, you do not need to install Python locally and you can store notebooks in Google Drive. You can also upload Jupyter notebooks inside CoLab. You can easily share CoLab notebooks just like a Google doc.

Jupyter Notebooks: I will use Jupyter notebooks to disseminate my lecture notes and homework problem sets. Jupyter notebooks are human-readable documents that contain code and text as well as other media like images and equations formatted in LaTeX. Notebook documents have an **.ipynb* extension and are created and viewed in a web browser. Jupyter Notebook comes bundled with the Anaconda distribution and can be launched from the Anaconda Navigator. Once it's launched, you can create new notebook documents or open existing ones. You may also want to check out Jupyterlab which is similar to Jupyter Notebook but has additional features and some feel it is easier to navigate. Apparently Jupyterlab may eventually overtake Jupyter Notebooks. More information about Jupyter Notebook and Jupyterlab is found at https://jupyter.org/. If you use an IDE like Spyder, you can also download code from notebook documents and load it easily into your IDE. Likewise, you can develop your code in your IDE and copy it into a notebook.

Help with Python: There are many Python manuals available, too many to list, and while I own a number of paper manuals, I find the quickest way to get Python help is with an internet search. Python is such a well-used language, chances are someone else has had your same question and has asked it somewhere online, oftentimes at stackoverflow.com. The wikipage for stackoverflow.com includes says this about it "*The site and similar programming question and answer sites have globally mostly replaced programming books for day-to-day programming reference in the 2000s, and today are an important part of computer programming*." In my internet search, I usually start with "python" then add keywords specific to my problem, e.g. "python numpy array initialize". Using Google, you can search a specific site like stackoverflow.com using "site:stackoverflow.com [keywords]". If you get an error message when running your code, you can often find out what it means by copying and pasting it directly into your browser's omnibar. In the IPython console on an IDE or in Jupyterlab, you can get documentation on a specific function using >>> *help(function_name),* or simply precede the function name with a question mark, >>> *? function_name*. This works only if the function is loaded into the current session. If you're using the Spyder IDE, you can also use the *Help* window. Of course, I'm happy to do my best to answer your Python questions too, but try the internet first. There is also lots of information at the official Python website, https://www.python.org/, and at websites for specific modules like https://numpy.org/ and matplotlib.org.

Class meetings: You are expected to read the assigned sections from the text before each class. I will review material and offer examples through Jupyter notebooks that will be posted on Canvas. The lecture notebooks will often contain missing code you will complete during class for practice. While you can use the computers in the classroom, you will likely want to bring your laptop to keep all of your work in one area. Completion of in-class coding exercises goes into your participation grade. See Grading Policy below.

Class Schedule: The following is a tentative class schedule of topics we will cover each day. It roughly follows the organization of topics in Newman, with some exceptions and some topics omitted.

Week	Tuesday	Thursday	HW due dates (by 11:59 pm)
1	**Mar 31** Python, Anaconda, Jupyter notebook, variables, arithmetic, built-in functions, modules, comments. Ch 1, §2.1-2.2.	**Apr 2** Control flow, *if-else*, *while loops*, *for* loops, lists, tuples, dictionaries. §2.3,2.4.1, 2.5.	
2	**Apr 7** NumPy, arrays, array arithmetic, reading data from files, user-defined functions. §2.4.2-2.4.5, 2.6.	**Apr 9** Matplotlib, plotting, drawing, animation, good programming style. Ch. 3, §2.7.	**HW1** – Apr 7
3	**Apr 14** Numerical error, program speed. Ch. 4.	**Apr 16** Numerical integration, trapezoidal rule, Simpson rule, approximation errors. §5.1-§5.3.	**HW2** –Apr 14
4	**Apr 21** Gaussian quadrature improper integrals, multiple integrals. §5.6-5.9.	**Apr 23** Numerical differentiation**,** second derivatives, partial derivatives, derivatives of measured data. §5.10 (skip §5.10.4-5.10.5).	**HW3** –Apr 21
5	**Apr 28** Solving nonlinear equations, relaxation method, binary search, Newton's method. §6.3.1-§6.3.6.	**Apr 30** Ordinary differential equations, Euler's method, Runge-Kutta methods. §8.1.1-§8.1.3.	**HW4** –Apr 28
6	**May 5** **MIDTERM**	**May 7** Second-order differential equations, ODEs with more than one variable. §8.2-§8.3.	**HW5** –May 5
7	**May 12** Adaptive step size, ODE boundary value methods. §8.4, §8.6.1-§8.6.2.	**May 14** Partial differential equations, finite differences, Jacobi method, Gauss-Seidel method, overrelaxation. §9.1-9.2.2.	**HW6** –May 12
8	**May 19** PDE initial value problems, FTCS method, numerical stability. §9.3.1-§9.3.2.	**May 21** Fourier series, Fourier transforms, discrete Fourier transforms. §7.1-§7.2.2.	**HW7** –May 19

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9	**May 26** Fast Fourier transforms, discrete cosine transforms. §7.3-7.4.	**May 28** Random numbers, seeds, probabilities, non-uniform and Gaussian random numbers. §10.1.1-§10.1.5	**HW8** – May 26
10	**June 2** Basic statistics, model fitting, curve fitting. §11.1-11.5.	**June 4** Project presentations.	**HW9** – June 2
11	**Term projects due** – Sunday, June 7, 11:59pm. **FINAL** – Thursday June 11, 11:10-1:00pm		

Homework (40%): Homework sets will give you ample opportunity to build your Python skills and solve physics problems using the numerical techniques we learn in class. Exercises will include writing Python code, physics and math derivations, and responses to questions about numerical techniques. Homework sets will be available to you in a Jupyter notebook on Canvas. Enter your code and solutions into the same notebook and submit your completed notebook to Canvas. For problems that require derivations and equations, you may enter your solutions directly into the notebook using embedded LaTeX, or embed a picture of your handwritten solutions. In either case, all of your solutions must be included inside your notebook.

Each homework set has a different total point value. Your homework grade is based on the percentage of the total points you earn. Homework will be due each Tuesday by 11:59pm. There is a 48-hr grace period on homework when homework is still accepted with a 10% deduction. After 48 hours no credit is given.

Note, many of the numerical methods we will use are available in Python modules or libraries. In most cases, I want you to learn how the numerical method works by programming it yourself first. After you get some practice with the method, you can import the function from the relevant module. If you are in doubt when it is appropriate to do so, please ask. Also, while there many modules you may find useful when completing homework problems *use only modules we have introduced in class for homework*, unless you have prior approval from me.

I encourage students to work together on homework, but the homework you submit must be your own.

Exams (35%): We will have two exams of equal weight, a midterm and a final, which will test understanding of course concepts, numerical techniques and coding. Exams will be administered in class. You will be given a formula sheet for each exam. The final exam covers only material after the midterm.

In-class practice (10%)**:** We will spend part of most classes working on coding exercises based on lecture materials. I encourage you to work on these practice exercises in partners or small groups using a shared Google CoLab notebook. Submit your completed notebook to the Canvas assignment folder at the end of each class session for participation credit. If working with others, only one member from your group needs to submit your notebook. Include names of those working with you. Your code does not need to work to receive credit, but it most show effort. You can miss 3 sessions with no loss in credit. I offer a “no-participation” grading option where the in-class practice weight is instead added to the exams grade.

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Term project (15%): The term project is an opportunity for you to apply Python to a topic of interest. This could be related to your current research, a numerical method we haven't covered in class, an interesting physics problem, or the development of an educational simulation like a Phet (https://phet.colorado.edu) using 3D graphics. Ph 316 students may wish to base their project on their Ph 316 experimental project. You will write your project in the format of a research paper and submit your paper, code and results in a Jupyter notebook. You can earn extra credit by giving a brief presentation of your project during the final week of classes. More details on the project are provided on Canvas.

Assessment

	Participation	**No participation**
Homework	40%	40%
Midterm	17.5	22.5
Final	17.5	22.5
In-class practice	10	0
Term project	15	15
Extra credit presentation	3	3

Use of AI**:** As you likely know, generative AI such as ChatGPT and Gemini are skilled coders. Most (all?) professional programmers now use AI to write much of their code, and some speculate in the not-so-distant future nearly all coding will be done with AI. So where does that leave us?

Well for starters, we are not in that future yet and humans are still writing code. Secondly, most of you will not be professional programmers, but will use Python and other programming languages as a means-to-an-end to conduct research and solve problems. It is still important you understand the fundamentals of programming and numerical techniques. You still need to verify and validate AI-generated code to guard against "hallucinations". You will still need to revise, edit and tailor AI-generated code for own purposes. You will still need to understand the fundamentals of numerical analysis and know the limits of numerical solutions and estimate uncertainty and accuracy. The process of learning to program and code builds your critical thinking skills, as it requires you to break complex problems down into smaller more manageable parts. Programming also requires you to think algorithmically. All of this is to say, that the skills you will learn in this class are still quite valuable even in this new age of AI.

Though you can use AI to help with homework, I strongly encourage you to write most of your code yourself, at least initially, when you are still learning Python and the numerical methods. Once you have mastered the fundamentals, it may make sense to use AI more and more. I encourage you to struggle with the homework problems, and only enlist AI when needed. AI can be an excellent tutor and debugger. If you get stuck, or don't know why your program isn't working, copy your code into the AI and ask for help. If you use AI-generated code in your homework or student, you must cite it. Make it clear in the comments of your program any code that was written by AI, and provide a citation along with the prompt you used. If in doubt, please ask me.

Grades: Grades will be assigned as follows:
A-range 90-100% of total points
B-range 80-89.9%

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C-range 70-79.9%
D-range 60-69.9%
F below 60
'+' for upper third of each range, i.e., B+ = 87-89.9%
'-' for lower third of each range, i.e. B- = 80-82.9%

Additional PSU info:

The CARE program connects students with resources when they are experiencing distress and/or other issues that might impact their ability to function at their best in and outside of the classroom such as mental health and medical challenges, and food and housing insecurity. CARE personnel meet with students one-on-one to hear about their academic barriers, and to collaborate on a plan to address these barriers in an effort to ease the stress of navigating multiple systems. You can learn more about CARE and schedule an appointment at https://www.pdx.edu/dean-student-life/care.

Academic Misconduct:
Academic dishonesty is the act of knowingly or intentionally seeking to claim credit for the work or effort of another person or participation in such acts. This includes, but is not limited to: (a) cheating, (b) fraud, (c) plagiarism, such as word for word copying, using borrowed words or phrases from original text into new patterns without attribution, or paraphrasing another writer's ideas; (d) The buying or selling of all or any portion of course assignments and research papers; (e) Performing academic assignments (including tests and examinations) for other persons; (f) Unauthorized disclosure or receipt of academic information; (g) Falsification of research data; and (h) Unauthorized collaboration.
Adapted from: PSU Student Code of Conduct (http://www.pdx.edu/dos/codeofconduct)
Cheating during an exam will result in a zero grade in the course and referral to the office of student affairs. A no tolerance policy will be enforced.

Title IX – Portland State is committed to providing an environment free of all forms of prohibited discrimination and sexual harassment (sexual assault, domestic and dating violence, and gender or sex-based harassment and stalking). If you have experienced any form of gender or sex-based discrimination or harassment, know that help and support are available. PSU has staff members trained to support survivors in navigating campus life, accessing health and counseling services, providing academic and on-housing accommodations, helping with legal protective orders, and more. Information about PSU's support services on campus, including confidential services and reporting options, can be found on PSU's Sexual Misconduct Prevention and Response website at: http://www.pdx.edu/sexual-assault/get-help or you may call a confidential IPV Advocate at 503-725-5672. Please be aware that all PSU faculty members and instructors **are required to report** information of an incident that may constitute prohibited discrimination, including sexual harassment and sexual violence. This means that if you tell me about a situation of sexual harassment or sexual violence that may have violated university policy or student code of conduct, I have to share the information with my supervisor or the University's Title IX Coordinator or the Office of Affirmative Action. For more information about Title IX please complete the required student module Creating a Safe Campus in your D2L.

Disability Accommodations at PSU – PSU values diversity and inclusion; we are committed to fostering mutual respect and full participation for all students. My goal is to create a learning environment that is equitable, useable, inclusive, and welcoming. If any aspects of instruction or course design result in barriers to your

inclusion or learning, please notify me. The Disability Resource Center (DRC) provides reasonable accommodations for students who encounter barriers in the learning environment. If you have, or think you may have, a disability that may affect your work in this class and feel you need accommodations, contact the Disability Resource Center to schedule an appointment and initiate a conversation about reasonable accommodations. The DRC is located in 116 Smith Memorial Student Union, 503-725-4150, drc@pdx.edu, https://www.pdx.edu/drc.

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